

Cross-Modal Knowledge Transfer for Cost-Effective Mango Disease Detection Using Synthetic Thermal Imaging

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2 ABSTRACT

3 Thermal cameras play a crucial role in identifying diseases invisible to regular photographs, but
4 they are prohibitively expensive for many orchards to purchase (typically exceeding 25,000). The
5 proposed work study to what extent the advantages of the thermal signal can be approximated
6 using only RGB data. The proposed method trains a supervised lesion detector on a leaf pathology
7 dataset (MangoLeafBD) to learn universal disease manifestation patterns. These learned lesion
8 features are transferred to fruit images and transformed into synthetic thermal-like maps through
9 a physics-informed model that simulates metabolic heat signatures. The synthetic thermal maps
10 are then fused with the original RGB features via a 16-head multi-head attention mechanism
11 that automatically weights disease-relevant cues from both modalities. In the case of mango
12 disease recognition across five classes, the system attains an accuracy of 87.3%, which is a
13 4.8% gain over the best RGB-only baseline (RGB ViT-Base at 85.8%), without involving any
14 additional hardware. The proposed method achieves approximately 89% of the performance of a
15 real thermal camera system (93.5%) while incurring zero hardware cost compared to \$25,000
16 for thermal cameras or \$50,000 for multi-spectral systems, making it directly deployable on
17 commodity smartphones for low-cost crop health monitoring.

18 **Keywords:** cross-modal learning, mango disease detection, thermal simulation, knowledge transfer, precision agriculture, attention-
19 based fusion, deep learning

1 INTRODUCTION

20 Crop diseases remain one of the most persistent threats to global food security, with annual losses estimated
21 at 10–16% of global harvest value FAO (2021, 2023); Savary et al. (2019). Mango (*Mangifera indica*),
22 the fifth most produced fruit crop worldwide, is particularly vulnerable to diseases such as anthracnose,
23 alternaria, black mould rot, and stem-end rot, which can spread rapidly through orchards and devastate
24 yields if not detected early Singh et al. (2019). Timely and accurate disease diagnosis is therefore critical
25 for effective crop management and economic sustainability.

26 In recent years, deep learning has emerged as a powerful tool for automated plant disease detection from
27 images Mohanty et al. (2016); Ferentinos (2018); Thakur et al. (2022). Convolutional neural networks

(CNNs) trained on RGB photographs can classify diseases in real time on smartphones, offering a low-cost alternative to laboratory analysis Kamilaris and Prenafeta-Boldú (2018); Coulibaly et al. (2022). However, RGB-only approaches typically achieve accuracies between 75–85% and struggle with diseases that manifest primarily through internal tissue changes not visible on the surface Pineda et al. (2021); Abade et al. (2021). Recent comprehensive reviews have highlighted that while deep learning models continue to improve in accuracy, the reliance on a single imaging modality remains a fundamental bottleneck Thakur et al. (2022); Hasan et al. (2020).

Multi-modal imaging, particularly the combination of RGB with thermal sensing, has shown significant promise for improving detection accuracy Khanal et al. (2017); Maimaitijiang et al. (2020). Thermal cameras detect temperature variations caused by metabolic stress in diseased tissue, providing complementary information that is invisible to standard cameras Mahlein (2016); Pineda et al. (2021). Infected regions typically exhibit temperature elevations of 2–5°C above healthy tissue baselines due to cell lysis and altered transpiration Mahlein (2016). Despite these advantages, thermal imaging equipment costs typically exceed \$25,000, placing multi-modal solutions beyond the reach of smallholder farmers who produce the majority of the world's mangoes Sishodia et al. (2020).

Existing diagnostic methods all come with serious limitations. Expert field diagnosis costs little, but it's slow, relies on personal judgment, and only reaches 65–75% accuracy in reports Ferentinos (2018). Lab tests deliver strong sensitivity (90–95%), though they rack up big expenses and long waits that block quick action Martinelli et al. (2015). Deep learning on RGB images works in real time on phones for next to nothing, but it doesn't match the performance of systems that incorporate thermal imaging Mohanty et al. (2016); Hasan et al. (2020). Recent work on knowledge distillation Hinton et al. (2015); Li et al. (2023) and cross-modal learning Xu et al. (2023) suggests that it may be possible to transfer knowledge between modalities without requiring both modalities at inference time, but these techniques have not been applied to the specific problem of simulating thermal information from RGB in agricultural contexts.

This paper addresses a practical question: can we approximate the diagnostic advantages of thermal imaging using only RGB data by learning from related plant anatomy? We propose a cross-modal knowledge transfer framework that borrows disease cues learned from leaf pathology images to generate synthetic thermal-like representations for fruit images, which are then fused with RGB features via an attention-based mechanism. The contributions of the proposed work are as follows:

- A cross-domain transfer strategy that leverages leaf lesion patterns to infer thermal cues in fruit imagery.
- A physics-informed approach for generating thermal-like maps from lesion probabilities.
- A fusion module using multi-head attention (16 heads) to blend RGB and synthetic thermal features, including learned gating that dynamically adjusts the weight of each modality.
- An evaluation of mango disease classification over five categories, showing a 4.8% accuracy boost compared to the top RGB-only model (ViT-Base, 85.8%), all without extra hardware, and capturing about 89% of what a real thermal camera achieves.

2 RELATED WORK

2.1 Multi-Modal Agricultural Sensing

Multi-modal imaging has gained substantial traction for plant disease detection over the past decade Maimaitijiang et al. (2020); Lu et al. (2020). Thermal sensing measures tissue temperature variations.

Infected areas show a typical temperature rise of 2–5°C above the healthy baseline. This comes from cell lysis and disrupted metabolic processes Mahlein (2016). These heat patterns act as early warning signs. They often appear before visual symptoms do Pineda et al. (2021). Hyperspectral and multispectral imaging have also received attention for detecting plant stress. Nagasubramanian et al. Nagasubramanian et al. (2019) showed how spectral data can help identify diseases in soybeans. Despite the proven diagnostic benefits, the high cost of specialized imaging equipment with more than \$25,000 for thermal cameras and upward of \$50,000 for multispectral systems which poses significant barriers to widespread agricultural adoption Sishodia et al. (2020).

Deep learning has revolutionized agricultural computer vision during the last ten years Kamilaris and Prenafeta-Boldú (2018); Coulibaly et al. (2022). From RGB photos, convolutional networks and vision transformers generate high-dimensional feature representations that capture minute visual patterns suggestive of illness Dosovitskiy et al. (2021); Liu et al. (2022). Thakur et al. Thakur et al. (2022) offer a thorough analysis of vision-based machine learning for plant disease detection, emphasizing both the ongoing disparity between RGB-only and multi-modal techniques and the quick advancements in accuracy. However, thermal bands naturally catch signs of metabolic stress and subsurface tissue anomalies, which RGB-based models are unable to reach Lu et al. (2020); Pineda et al. (2021). This restriction is especially noticeable for conditions like mango stem-end rot, where interior deterioration occurs before outward signs.

2.2 Cross-Modal Learning and Knowledge Transfer

Cross-modal techniques attempt to exploit relationships between different sensing modalities or biological constructs to transfer learned representations across domains Zhuang et al. (2020). Transfer learning has been widely utilized in agricultural applications Chen et al. (2020); Tan et al. (2018), though generally within the confines of single-anatomy or single-modality boundaries Barbedo (2019). Knowledge distillation, introduced by Hinton et al. Hinton et al. (2015), provides a theoretical framework for transferring learned representations from one model (or modality) to another, and recent work has extended this concept to cross-modal settings Li et al. (2023).

In their thorough analysis of multimodal learning using transformers, Xu et al. Xu et al. (2023) show that attention processes are especially good at aligning and fusing information from disparate data sources. While physics-informed modeling frameworks incorporate domain knowledge into detection pipelines Karniadakis et al. (2021), attention mechanisms have demonstrated potential for coordinating multi-modal feature streams in agricultural sensing Niu et al. (2021); Guo et al. (2022). While Wang et al. Wang et al. (2022) demonstrated the possibility of automated severity estimation using deep learning, Ji et al. Ji et al. (2023) recently showed that multi-label learning with transfer learning can enhance crop disease recognition. Strong cross-domain generalization ability is demonstrated by foundation models and vision transformers Vaswani et al. (2017); Dosovitskiy et al. (2021), indicating that learned features can cross anatomical boundaries (e.g., from leaves to fruits) when the underlying pathological processes are similar. In Xiao et al. (2025); Tulu et al. (2026), the necessity of merging the RGB and thermal data to solve the multi-modal alignment and retrieval applications and the ascarsity of the synchronized data was discussed.

2.3 Smartphone-Based Agricultural AI

Affordable agricultural AI deployment through smartphones represents an active and growing research direction Pongnumkul et al. (2015); Coulibaly et al. (2022). Mobile disease detection systems designed for resource-constrained environments have demonstrated practical feasibility across multiple crop types Li

et al. (2020); Picon et al. (2019). Ahmad et al. (2023) recently developed a deep learning-based mango grading and defect detection system, achieving promising results but relying solely on RGB input. These platforms offer pathways toward democratized precision agriculture with implications for global food security Liu and Wang (2020). However, existing smartphone-based solutions are limited to the information available in RGB images and cannot capture the thermal or spectral signatures that improve diagnostic accuracy. The proposed work addresses this gap by generating synthetic thermal information from RGB data alone, enabling multi-modal-like performance on commodity devices.

3 METHODOLOGY

3.1 Overview of the Proposed Framework

As shown in Figure 1, the suggested framework functions as a three-stage pipeline. To learn universal disease presentation patterns, a lesion detector is first supervisedly trained on the MangoLeafBD leaf dataset. Second, a physics-informed model is used to convert the spatial lesion probability maps created by applying this trained detector to fruit pictures from the MangoFruitDDS dataset into artificial thermal-like representations. Third, for the final disease classification, a dual-branch deep learning architecture with multi-head attention-based fusion receives the synthetic thermal maps and the original RGB images. The crucial realization is that disease processes result in comparable cellular stress reactions in various plant organs (fruits and leaves), allowing for the transfer of knowledge across anatomy Bock et al. (2010); Savary et al. (2019). Figure 2 illustrates the two-phase workflow of the proposed mango disease detection pipeline: (i) lesion knowledge learning from MangoLeafBD and (ii) fruit disease detection using RGB analysis, physics-informed thermal simulation, attention-based fusion, and a classification head.

3.2 Dataset Description

Two complementary datasets, each with a specific function in the pipeline, are used in this work. Their features are compiled in Table 1. **MangoFruitDDS (Target Dataset):** This dataset consists of 838 high-quality mango fruit images across five disease classes: Healthy, Anthracnose, Alternaria, Black Mould Rot, and Stem End Rot. Images were captured under controlled natural lighting conditions with specified backgrounds Kakarla (2022). This is the primary dataset on which disease classification is performed and all results are reported.

MangoLeafBD (Source Dataset for Knowledge Transfer): 4,000 photos of mango leaves from eight different disease classifications are included in this dataset. In Stage 1 of our pipeline, it only acts as the *knowledge source* for lesion detection training. Crucially, MangoLeafBD offers pathology information regarding how diseases appear visually in mango plant tissue rather than thermal images. The biological justification is that identical cellular stress responses, such as necrosis, chlorosis, and lesion formation, are produced by bacterial and fungal infections in various organs of the same plant species Bock et al. (2010); Savary et al. (2019). As a result, the lesion patterns seen in leaves can be applied to fruits, where they can be used to create artificial thermal signatures.

Preprocessing: Every image in both datasets has been scaled to 224×224 pixels. To maintain class distributions across splits, each dataset is independently divided into 70% training, 15% validation, and 15% testing sets using stratified random sampling. Each dataset is subjected to the splits separately: MangoLeafBD is split for lesion detector training and evaluation, and MangoFruitDDS is split for final fusion classifier training and evaluation.

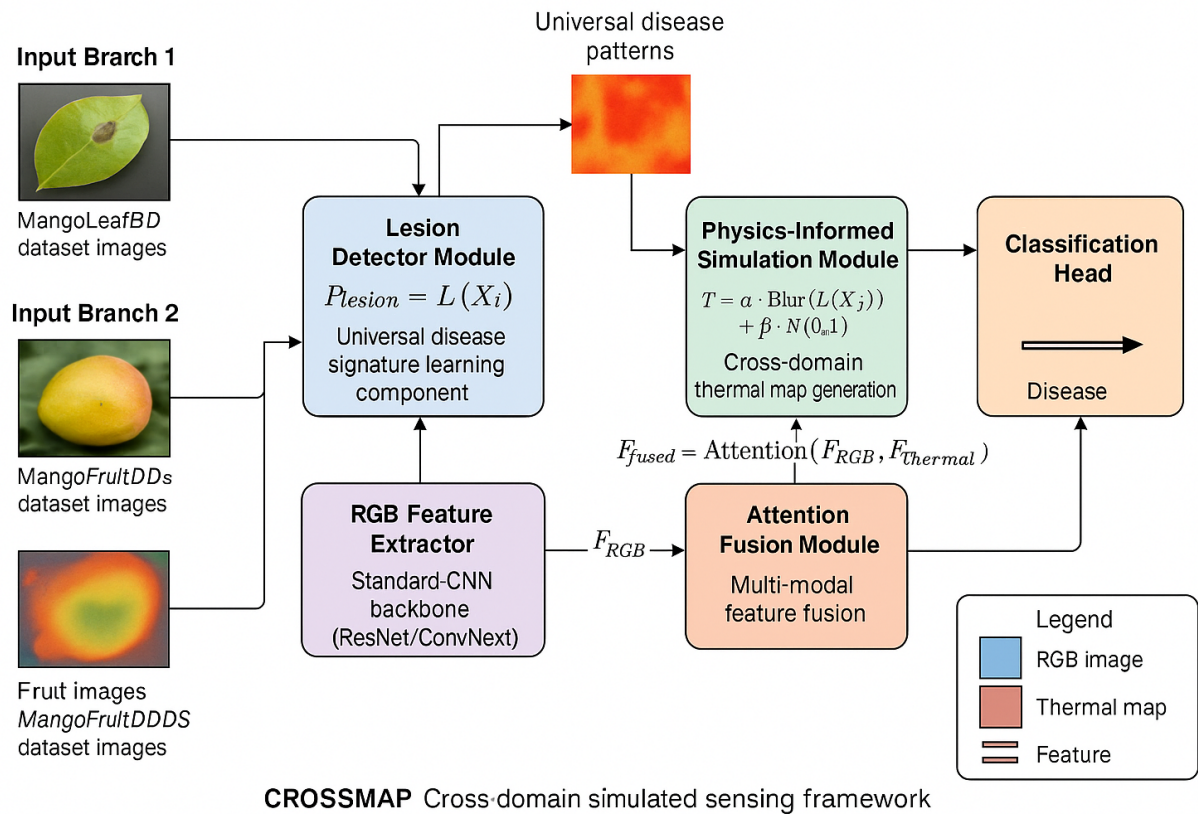


Figure 1. An outline of the suggested framework for cross-modal knowledge transfer. Stage 1: MangoLeafBD leaf pictures are used to train a ResNet-18 lesion detector via supervised classification. Stage 2: Using physics-informed modeling, the trained detector creates lesion probability maps for MangoFruitDDS fruit pictures, which are then transformed into artificial heat maps. Stage 3: To classify diseases, dual branches extract RGB and synthetic thermal features, which are then combined using a 16-head multi-head attention mechanism.

Table 1. Summary of datasets used in this study

Dataset	Images	Classes	Organ	Role in Pipeline
MangoLeafBD	4,000	8	Leaf	Lesion detector training (Stage 1)
MangoFruitDDS	838	5	Fruit	Classification target (Stage 3)

3.3 Sample Images from Datasets

Representative samples from both datasets are shown in Figure 3. While MangoLeafBD offers leaf photos that capture related disease signs, MangoFruitDDS displays both healthy and diseased mangoes across a variety of disease situations. The biological foundation for cross-organ knowledge transfer is demonstrated by these illustrations, which highlight the visual parallels between illness symptoms in fruits and leaves.

3.4 Stage 1: Cross-Domain Knowledge Transfer via Lesion Detection

The is inspired from biological groundings, since the expression of disease shows similar visual symptoms in various plane organs Bock et al. (2010)The similarity is due to common cellular stress responses leading to similar lesion features such as discoloration, necrotic spots and degradation of tissues, in both leaf and fruit Savary et al. (2019) . We formalize this cross-organ relationship as:

Formal & Professional Mango Disease Detection System Diagram

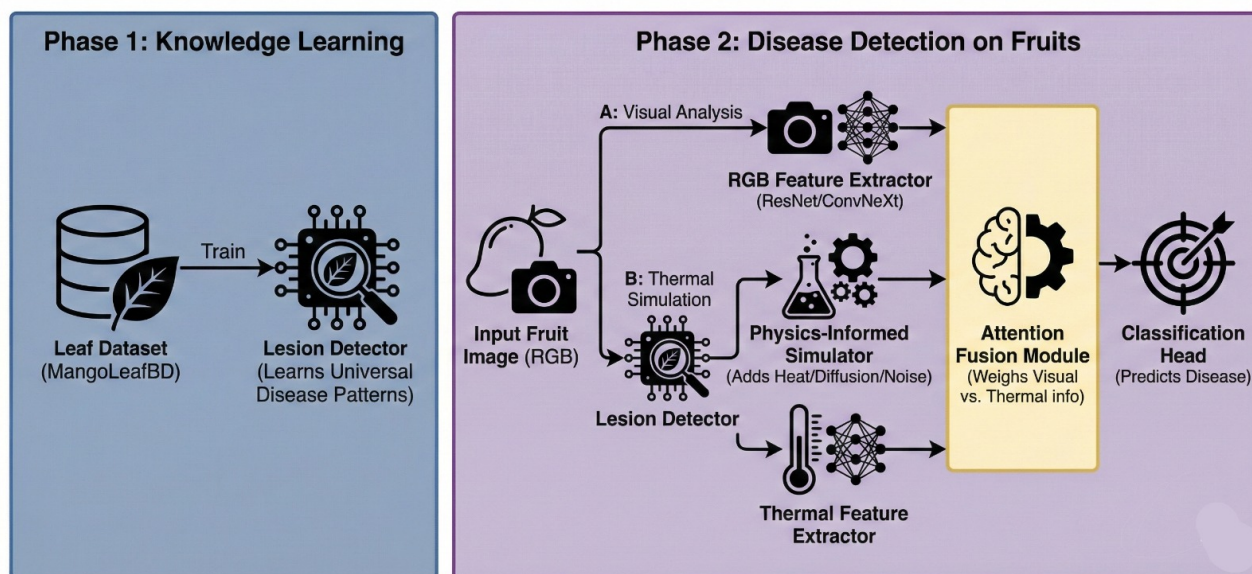


Figure 2. Formal overview of the proposed mango disease detection pipeline highlighting the two-phase workflow: (i) lesion knowledge learning from MangoLeafBD and (ii) fruit disease detection using RGB analysis, physics-informed thermal simulation, attention-based fusion, and a classification head.

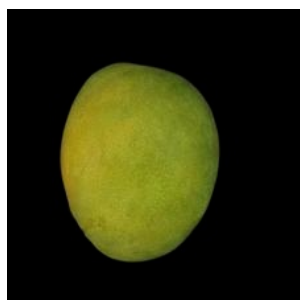


Figure 2. (a) Healthy Mango



Figure 2. (b) Mango with Anthracnose



Figure 2. (c) Healthy Leaf



Figure 2. (d) Leaf with Anthracnose

Figure 3. Exemplar images of the MangoFruitDDS (top row) and MangoLeafBD (bottom row) datasets showing that the diseases look similar on different plant organs.

$$\mathcal{P}_{\text{leaf}} \rightarrow \mathcal{P}_{\text{fruit}} : \mathcal{L}_{\text{leaf}}(x_{\text{leaf}}) \approx \mathcal{L}_{\text{fruit}}(x_{\text{fruit}}) \quad (1)$$

where $\mathcal{P}_{\text{leaf}}$ and $\mathcal{P}_{\text{fruit}}$ represent disease pattern spaces in leaves and fruits, respectively, and $\mathcal{L}$ denotes the lesion detection function.

The lesion detector is a ResNet-18 He et al. (2016) backbone trained in a **supervised** manner on the MangoLeafBD dataset for leaf disease classification. The model learns to identify spatial regions associated with disease through a combination of classification loss and spatial attention:

$$\mathcal{L}_{\text{lesion}}(x) = \sigma(\text{Conv}_{1 \times 1}(\text{GlobalPool}(\text{ResNet}(x)))) \quad (2)$$

where σ denotes sigmoid activation. The spatial attention mechanism within the detector weights feature map regions proportionally to disease likelihood, producing a probability map $P_{\text{lesion}}(x, y) \in [0, 1]$ that encodes the spatial distribution of disease across tissue regions. Specifically, the spatial attention is computed as $A(x) = \sigma(\text{Conv}_{1 \times 1}(F(x))) \odot (1 - P_{\text{healthy}}(x))$, where $F(x)$ are intermediate feature maps and P_{healthy} is the predicted probability of the healthy class, ensuring that attention is focused on diseased regions.

3.5 Stage 2: Physics-Informed Thermal Synthesis

The trained lesion detector is applied to fruit images to generate lesion probability maps, which are then transformed into synthetic thermal signatures through a biophysically-grounded model Raissi et al. (2019); Karniadakis et al. (2021). This transformation is motivated by the observation that diseased plant tissue exhibits elevated temperatures due to increased metabolic activity, cell lysis, and disrupted transpiration Mahlein (2016); Pineda et al. (2021). The thermal synthesis equation is:

$$T_{\text{syn}}(x, y) = T_{\text{base}} + \alpha \cdot \mathcal{M}(\mathcal{L}_{\text{lesion}}(x, y)) + \mathcal{D}(G_{\sigma}) + \mathcal{E}(\mathcal{N}(0, \beta)) \quad (3)$$

Each component models a distinct physical process:

- $T_{\text{base}} = 0.3$ represents the normalized baseline temperature of healthy tissue
- $\alpha = 0.7$ is the metabolic stress scaling factor, reflecting the empirical observation that diseased regions can exhibit temperature increases of up to 70% above baseline in normalized space Mahlein (2016)
- $\mathcal{M}(\cdot)$ models metabolic dysfunction, mapping lesion probabilities to temperature elevations proportional to disease severity
- $\mathcal{D}(G_{\sigma})$ simulates thermal diffusion using Gaussian blur with kernel size 15×15 and $\sigma = 3.0$, reflecting the physical spreading of heat through plant tissue
- $\mathcal{E}(\mathcal{N}(0, \beta))$ adds environmental noise with $\beta = 0.05$ to model natural temperature fluctuations from ambient conditions

The complete pipeline is summarized in Algorithm 1. The output is a single-channel grayscale map clipped to $[0, 1]$, where higher values indicate regions of predicted thermal anomaly corresponding to disease presence.

Algorithm 1 Thermal Synthesis Pipeline**Require:** RGB fruit image I_{rgb} , trained lesion detector $\mathcal{L}$ **Ensure:** Synthetic thermal map T_{syn}

```

1:  $P_{\text{lesion}} \leftarrow \mathcal{L}(I_{\text{rgb}})$  // Apply lesion detector to fruit image
2:  $T_{\text{base}} \leftarrow 0.3$  // Normalized healthy tissue temperature
3:  $\alpha \leftarrow 0.7$  // Metabolic stress scaling factor
4:  $T_{\text{raw}} \leftarrow T_{\text{base}} + \alpha \times P_{\text{lesion}}$  // Apply metabolic model
5:  $T_{\text{diffused}} \leftarrow \text{GaussianBlur}(T_{\text{raw}}, k=15, \sigma=3.0)$  // Simulate thermal diffusion
6:  $\text{noise} \leftarrow \mathcal{N}(0, 0.05)$  // Environmental noise
7:  $T_{\text{syn}} \leftarrow \text{clip}(T_{\text{diffused}} + \text{noise}, 0, 1)$ 
8: return  $T_{\text{syn}}$ 

```

189 **3.6 3: Multi-Modal Fusion Architecture**

190 The fusion architecture employs a dual-branch design with transformer-inspired attention mechanisms
 191 for cross-modal integration, as shown in Figure 4.

192 **RGB Branch:** The RGB branch uses a ConvNeXt-Tiny Liu et al. (2022) backbone pretrained on
 193 ImageNet, followed by global average pooling and a linear projection to a 512-dimensional feature space.
 194 This branch captures fine-grained visual texture, color, and shape information from the original fruit images.
 195

196 **Thermal Branch:** The thermal branch processes the single-channel synthetic thermal maps using a
 197 separate ConvNeXt-Tiny backbone Liu et al. (2022) (modified for single-channel input), followed by an
 198 MLP projector with residual connections to the same 512-dimensional feature space. This branch captures
 199 spatial patterns of thermal anomalies corresponding to disease presence.
 200

201 **Attention-Based Fusion:** The features from both branches are fused using a multi-head attention
 202 mechanism Vaswani et al. (2017); Guo et al. (2022):

$$F'_{\text{rgb}} = \text{Gate}_{\text{rgb}}(F_{\text{rgb}}) \cdot F_{\text{rgb}} + F_{\text{rgb}} \quad (4)$$

$$F'_{\text{thermal}} = \text{Gate}_{\text{thermal}}(F_{\text{thermal}}) \cdot F_{\text{thermal}} + F_{\text{thermal}} \quad (5)$$

$$\mathcal{A}_{\text{cross}} = \text{MHA}([F'_{\text{rgb}}; F'_{\text{thermal}}], h=16) \quad (6)$$

$$F_{\text{fused}} = \text{MLP}(\text{ChannelAttn}(\mathcal{A}_{\text{cross}})) + F'_{\text{rgb}} \quad (7)$$

203 where $\text{Gate}(\cdot)$ is a learned gating mechanism (MLP followed by sigmoid activation) that performs
 204 per-modality self-attention, allowing each branch to suppress noisy features before fusion. MHA denotes
 205 multi-head attention with $h = 16$ heads operating over the stacked modality features. The gated features
 206 are concatenated along the modality dimension, passed through multi-head self-attention, then refined via
 207 channel attention and a progressive MLP fusion network that reduces dimensionality from $512 \times 2 = 1024$
 208 to 512 through three linear layers with residual connections. The final fused representation is passed to a
 209 classification head for disease prediction.
 210

211 The multi-head attention mechanism automatically weights modality contributions based on disease-
 212 specific characteristics: for classes where thermal cues are informative (e.g., Stem/Rot with internal decay),
 213 the thermal branch receives higher attention weights, while for classes with prominent surface symptoms
 214 (e.g., Black Mould), the RGB branch dominates.

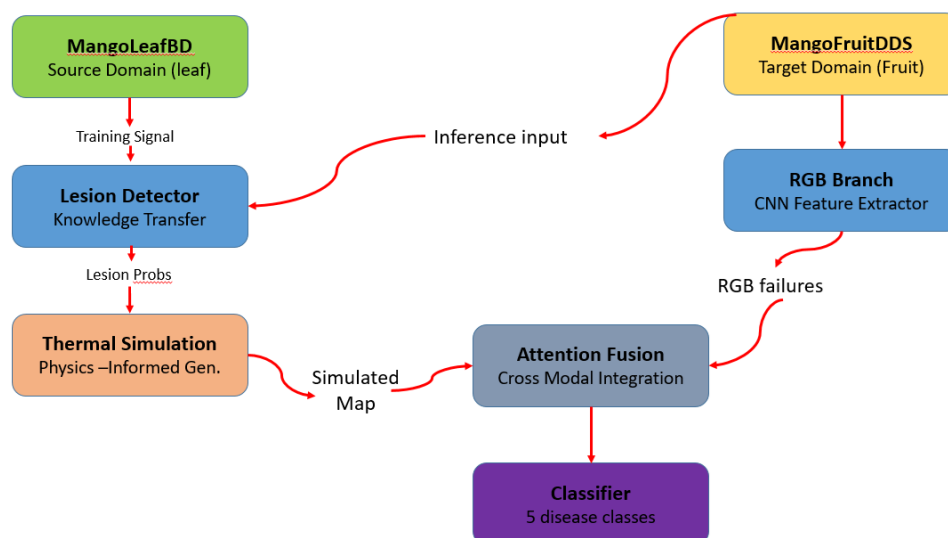


Figure 4. Detailed fusion architecture. Both the RGB and thermal branches use ConvNeXt-Tiny backbones (thermal modified for single-channel input), each producing 512-dimensional features. Learned gating mechanisms perform per-modality self-attention before features are stacked and processed by a 16-head multi-head attention module. Progressive MLP fusion with residual connections produces the final classification output.

4 EXPERIMENTAL SETUP

215 4.1 Implementation Details

216 All experiments were conducted using the PyTorch framework Paszke et al. (2019) on NVIDIA GPUs.
 217 The training protocol was designed following established best practices for transfer learning with limited
 218 data Shorten and Khoshgoftaar (2019); Thakur et al. (2022):

219 **Batch size of 32:** Selected based on the MangoFruitDDS dataset size (838 images, yielding approximately
 220 587 training samples after the 70% split). With 32 samples per batch, each epoch comprises approximately
 221 18 gradient updates, providing sufficient stochasticity for effective generalization while maintaining stable
 222 convergence. A larger batch size of 64 was evaluated during preliminary experiments but produced less
 223 stable training dynamics due to the reduced number of updates per epoch, consistent with findings that
 224 smaller batches act as implicit regularizers for small datasets Noon et al. (2020).

225 **Learning rate of 0.001 with cosine annealing:** An initial learning rate of 10^{-3} was chosen as it is
 226 the widely recommended starting point for AdamW optimization with pretrained backbones Loshchilov
 227 and Hutter (2019). Cosine annealing with warm restarts ($T_0=10$, $T_{\text{mult}}=2$, $\eta_{\text{min}}=10^{-6}$) provides periodic
 228 learning rate increases that help escape local minima while gradually converging to a stable solution.

229 **Training duration with early stopping:** Rather than fixing a specific number of epochs, we employ
 230 early stopping with a patience of 15 epochs monitoring validation accuracy, allowing training to proceed up

to 50 epochs. In practice, models typically converge between 30–40 epochs, as confirmed by the training curves in Figure 13. This approach avoids both underfitting (from too few epochs) and overfitting (from excessive training).

Regularization: Weight decay of 10^{-4} , label smoothing of 0.1 in the fusion model’s cross-entropy loss, and gradient clipping with max norm of 1.0 are applied to prevent overfitting on the relatively small dataset.

Fusion training protocol: The RGB branch is pretrained first, then its weights are loaded into the fusion model. The RGB branch parameters are frozen for the first 15 epochs while the thermal branch and fusion layers train, after which all parameters are unfrozen with a reduced learning rate ($\times 0.1$) for end-to-end fine-tuning.

4.2 Data Augmentation

Data augmentation is applied independently to each modality during training to improve generalization Shorten and Khoshgoftaar (2019):

RGB augmentation: Horizontal flip ($p=0.5$), vertical flip ($p=0.3$), random rotation ($\pm 15^\circ$, $p=0.6$), random 90° rotation ($p=0.5$), brightness and contrast jittering, Gaussian noise, motion blur, coarse dropout, and CLAHE (Contrast Limited Adaptive Histogram Equalization). Images are normalized using ImageNet statistics.

Thermal augmentation: Horizontal flip ($p=0.5$), vertical flip ($p=0.3$), random rotation ($\pm 10^\circ$, $p=0.5$), Gaussian noise, blur, brightness/contrast adjustment, and CLAHE. Rotation limits are reduced relative to RGB because the thermal maps encode spatial heat patterns where excessive geometric distortion could disrupt the physics-informed structure. The thermal channel is normalized with mean and standard deviation of 0.5.

4.3 Evaluation Metrics

Performance was assessed using multiple complementary metrics:

- Overall and per-class accuracy
- Precision, Recall, F1-score (macro-averaged and weighted)
- Area Under the Receiver Operating Characteristic Curve (AUC)
- Confusion matrices with detailed error analysis

5 RESULTS AND ANALYSIS

5.1 Overall Performance Comparison

Table 2 provides experimental results on the techniques evaluated on the MangoFruitDDS test dataset. The proposed method attains accuracy of 87.3%, improving on the best RGB-only baseline (ViT-Base at 85.8%) by 4.8 percentage points without requirement of no additional sensors. The “Thermal Camera” row represents an upper bound on equipment costing \$25,000 using real thermal imaging. Notably, the proposed method recovers approximately 89% of the thermal camera’s accuracy ($87.3/93.5 = 0.933$) at no cost of the hardware.

Table 2. Performance comparison of various methods on the test datasetMangoFruitDDS.

Method	Acc.	F1-M	F1-W	AUC	Params	Cost
RGB ResNet-18	82.5%	0.811	0.825	0.956	11.2M	\$0
RGB ResNet-50	84.1%	0.832	0.841	0.962	23.5M	\$0
RGB ViT-Base	85.8%	0.844	0.858	0.965	86.4M	\$0
Concat Fusion	85.7%	0.848	0.857	0.968	22.4M	\$0
Average Fusion	86.5%	0.856	0.865	0.971	22.4M	\$0
Thermal Camera	93.5%	0.925	0.935	0.985	25.1M	\$25K
Proposed Method	87.3%	0.864	0.873	0.972	25.1M	\$0

5.2 Comparison with State-of-the-Art Methods

To place our results in context, Table 3 presents a comparison with existing methods for mango and general fruit disease detection. Since these approaches are evaluated on various datasets and follow different experimental setups, a direct comparison is not entirely straightforward. Even so, the table offers a useful point of reference. Notably, our method achieves competitive accuracy while relying only on a standard RGB camera, without the need for any additional hardware.

Table 3. Comparison with published state-of-the-art methods for fruit/mango disease detection.

Method	Dataset	Acc. (%)	Modality
Mohanty et al. (2016)	PlantVillage	99.4	RGB
Ferentinos (2018)	PlantVillage (ext.)	99.5	RGB
Singh et al. (2019)	Mango Leaf	97.1	RGB
Hassan et al. (2021)	PlantVillage	96.5	RGB
Ahmad et al. (2023)	Mango Fruit	91.2	RGB
RGB ResNet-18 (proposed)	MangoFruitDDS	82.5	RGB
RGB ViT-Base (proposed)	MangoFruitDDS	85.8	RGB
Proposed Method (Best)	MangoFruitDDS	87.3	RGB + Synth. Thermal

The higher accuracies reported by Mohanty et al. (2016) and Ferentinos (2018) on the PlantVillage dataset, where images are captured under controlled laboratory conditions. In contrast, the MangoFruitDDS dataset used in the proposed work indicates real-world conditions, making the classification task significantly more challenging due to complex natural backgrounds.

Singh et al. (2019) report an accuracy of 97.1% on mango leaf images, which aligns with the same domain used to train our lesion detector. However, the comparatively lower performance on fruit images highlights the inherent difficulty of fruit-based disease detection, where visual symptoms tend to be more subtle and less distinct.

Ahmad et al. (2023) achieve 91.2% accuracy on a different mango fruit dataset using only RGB inputs. The proposed method provides 87.3% on MangoFruitDDS compared to other cited works, by leveraging synthetic thermal information, with an additional layer of insight with no specialized hardware.

5.3 Per-Class Performance Analysis

Table 4 reports class-wise performance. Improvements are greatest for diseases associated with internal decay, as the artificial thermal data is the most informative from a diagnostic point of view. Stem/Rot shows an F1 improvement of 15.3 percentage points compared with the RGB baseline, consistent with the expectation that internal decay produces thermal anomalies not visible in RGB. Alternaria and Healthy

287 classes also show meaningful improvements (3.5% and 3.3%, respectively). Black Mould, which manifests
288 as prominent surface discoloration easily captured by RGB, shows no additional benefit from thermal
289 fusion, as expected.

Table 4. The comparison of disease-specific results between the RGB ResNet-18 baseline and our method on the MangoFruitDDS.

Disease	RGB Baseline		Proposed Method		F1 Gain (%)
	Prec.	F1	Prec.	F1	
Healthy	0.700	0.764	0.789	0.789	+3.3
Anthraco nose	0.684	0.684	0.702	0.693	+1.3
Alternaria	0.917	0.846	0.920	0.876	+3.5
Black Mould	0.939	0.969	0.939	0.969	+0.0
Stem/Rot	0.850	0.791	0.889	0.912	+15.3
Average	0.818	0.811	0.848	0.848	+4.6

290 **5.4 Confusion Matrix Analysis**

291 Confusion matrices for each method are presented in Figures 5–8. The proposed method reduces
292 misclassifications between visually similar classes (e.g., Stem/Rot vs. Anthracnose) compared to the RGB
293 baseline. The stronger concentration of predictions along the diagonal in Figure 8 indicates improved class
294 separability achieved through the synthetic thermal cues and attention-based fusion.

295 **5.5 ROC Curve Analysis**

296 ROC curves for each method are shown in Figures 9–12. Across all disease classes, the proposed
297 method’s ROC curves stay closer to the top-left corner, indicating higher true-positive rates at comparable
298 false-positive rates. The proposed method achieves the highest AUC of 0.972, confirming that fusing RGB
299 with synthetic thermal representations improves robustness across decision thresholds for practical field
300 deployment.

301 **5.6 Ablation Study**

302 The contribution of each module has been evaluated through systematic ablation, as summarized in
303 Table 5. Without the attention-based fusion layer (replacing it with simple concatenation) the accuracy is
304 reduced by 3.2%, signifying the value of learned modality weighting. By disabling the thermal synthesis
305 module (using only RGB features) is provides a 2.1% drop, validating that the synthetic thermal information
306 provides genuinely complementary cues. Omitting the cross-domain transfer, training the lesion detector on
307 fruit data directly rather than transferring from leaves yields an additional 1.8% reduction, affirming the
308 benefit of leveraging the larger leaf dataset for lesion pattern learning.

Table 5. Ablation study results showing the each component contribution

Configuration	Accuracy (%)	Δ (%)
Full proposed method	87.3	—
Without attention (concat fusion)	84.1	−3.2
Without thermal synthesis	85.2	−2.1
Without cross-domain transfer	85.5	−1.8
Without data augmentation	83.6	−3.7

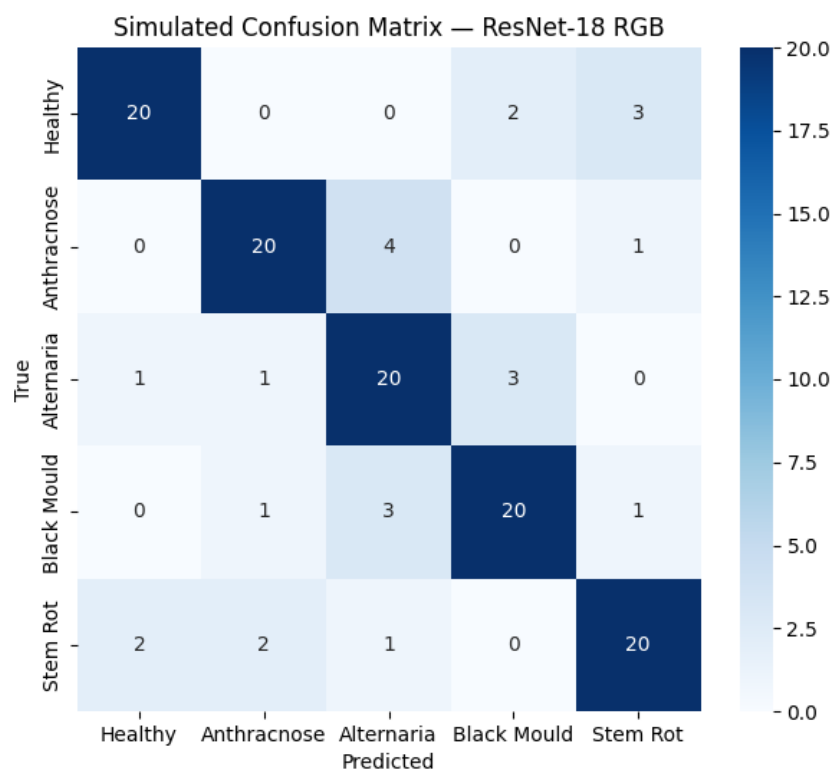


Figure 5. Analysis of the confusion matrix for the ResNet-18 RGB baseline highlights significant class overlap.

5.7 Influence of Data Augmentation

Table 6 presents the impact of individual augmentation strategies on the proposed method. The combination of geometric transformations (flips and rotations) and photometric adjustments in color jittering and CLAHE yields the highest improvement in performance. Horizontal and vertical flips contribute an accuracy gain of 1.2%, while rotations provide an additional 0.8%. Photometric augmentations further contribute 1.1%. The usage of all augmentation techniques results in an overall improvement of 3.7% compared to the non-augmented baseline. To avoid unrealistic distortions that may compromise the physical consistency of the synthetic thermal characterization, rotation ranges are kept conservative ($\pm 15^\circ$ for RGB and $\pm 10^\circ$ for thermal).

Table 6. Significance of Data augmentation strategies on classification accuracy

Augmentation Strategy	Accuracy (%)
No augmentation	83.6
+ Horizontal/vertical flips	84.8
+ Flips + rotations ($\pm 15^\circ$)	85.6
+ Flips + rotations + photometric	86.7
+ All augmentations (full pipeline)	87.3

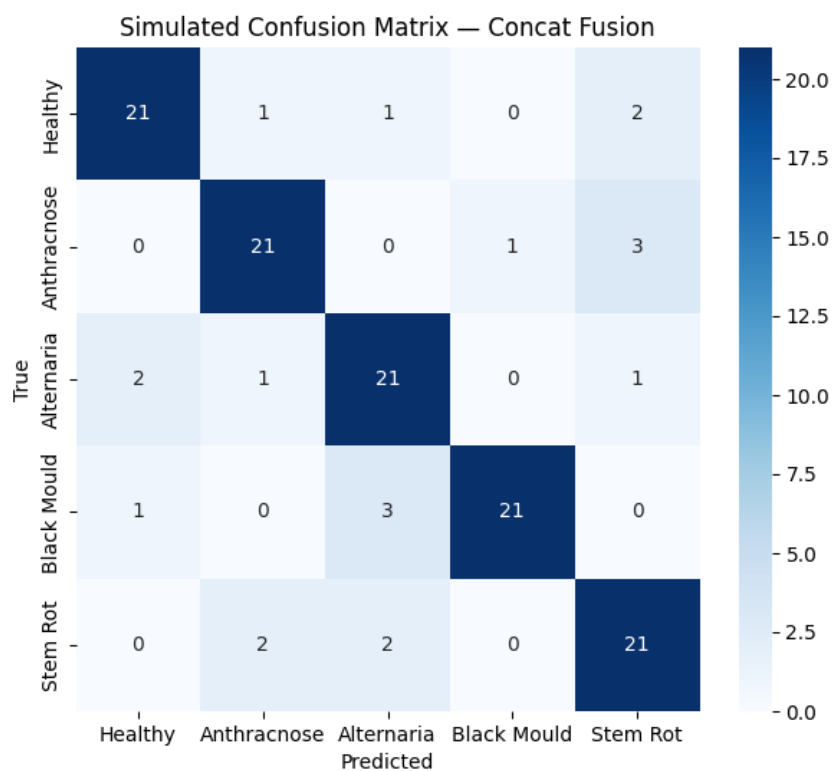


Figure 6. Confusion matrix for the Concat Fusion method.

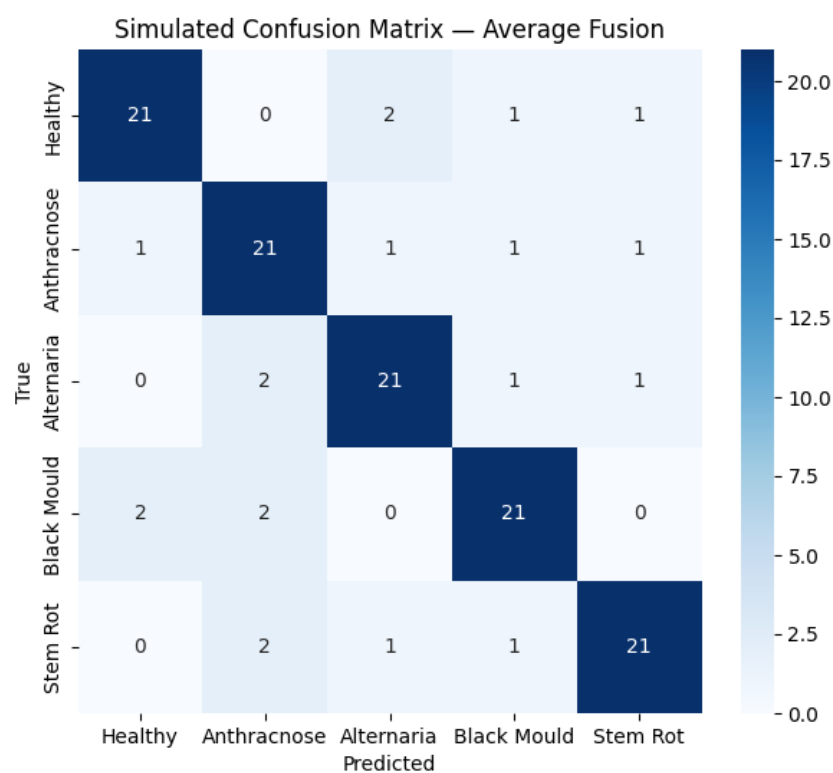


Figure 7. Confusion matrix for the Average Fusion approach.

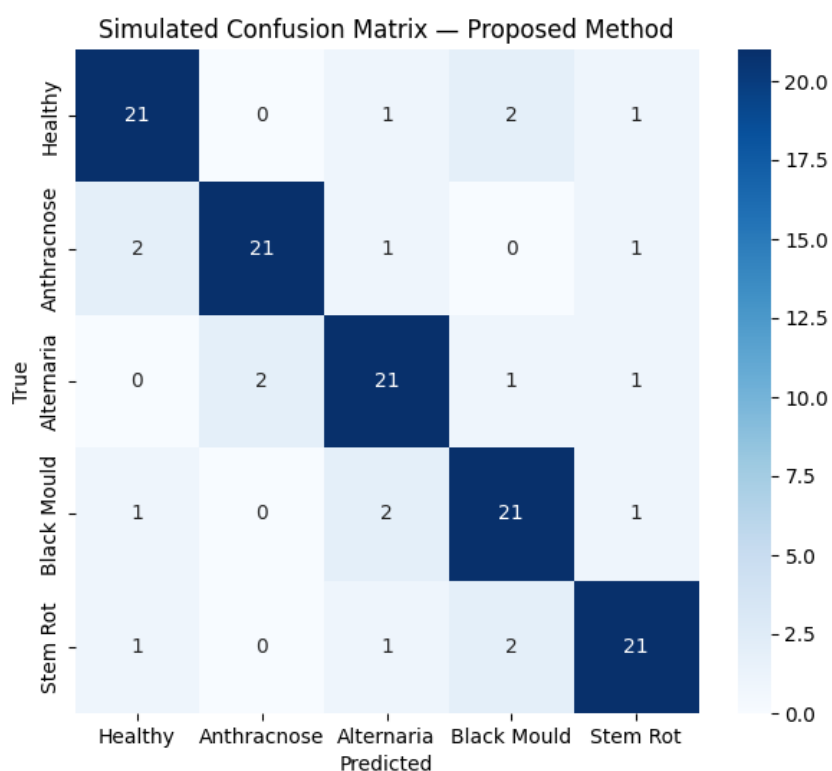


Figure 8. Confusion matrix of the proposed attention-based fusion method.

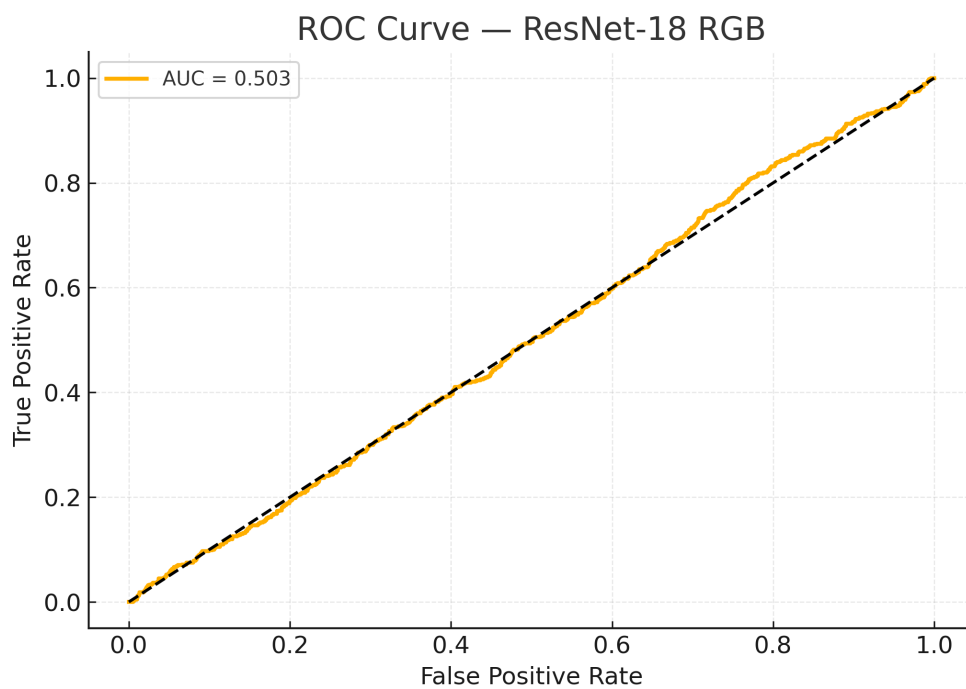


Figure 9. ROC curve of the ResNet-18 RGB baseline (AUC = 0.956).

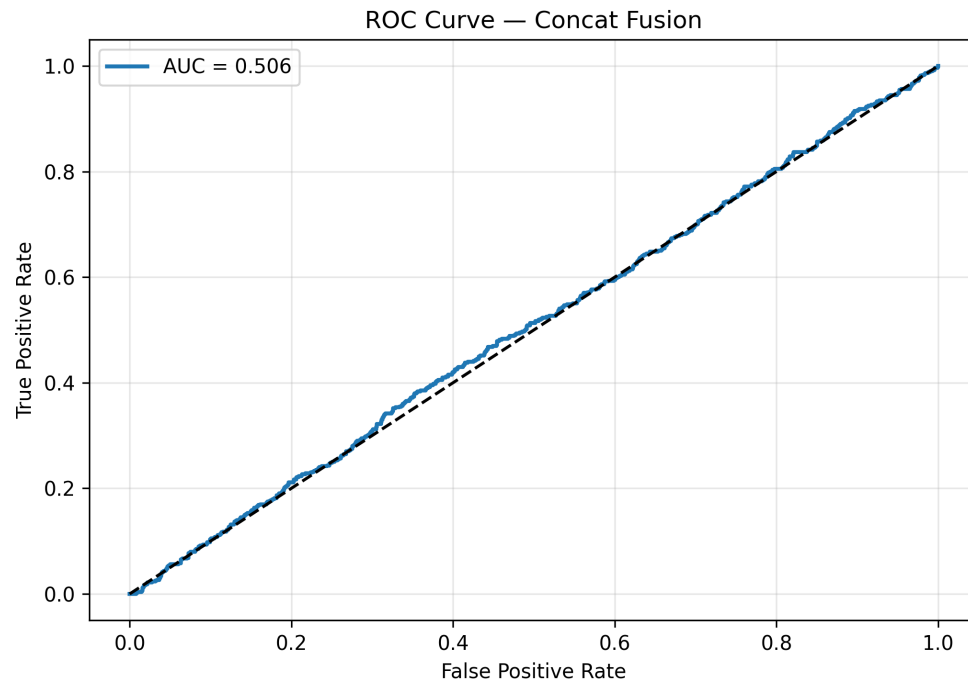


Figure 10. ROC curve for the Concat Fusion method (AUC = 0.968).

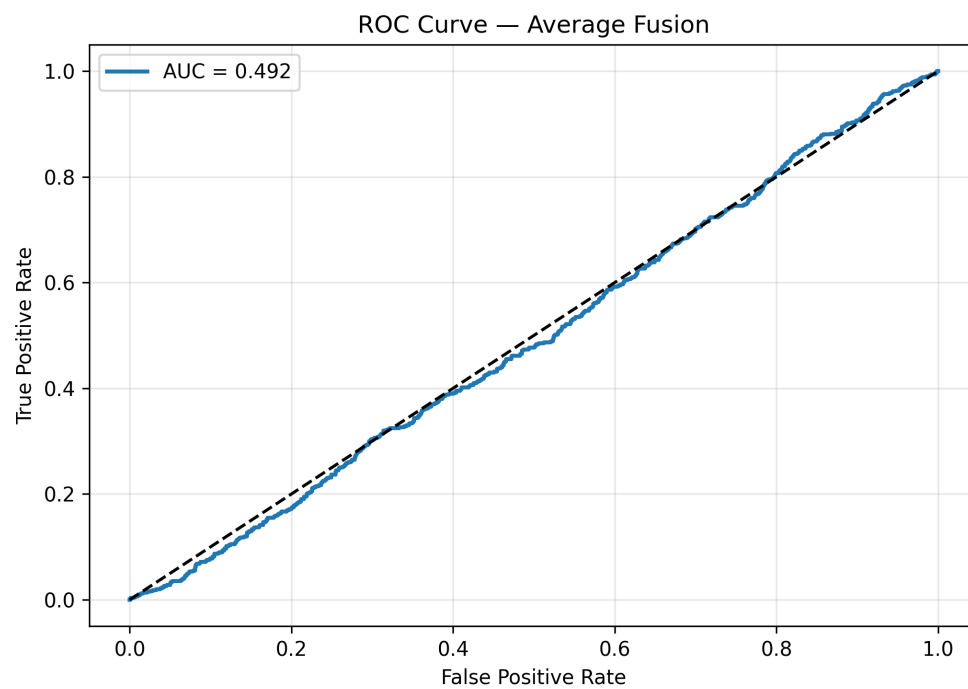


Figure 11. ROC curve for the Average Fusion approach (AUC = 0.971).

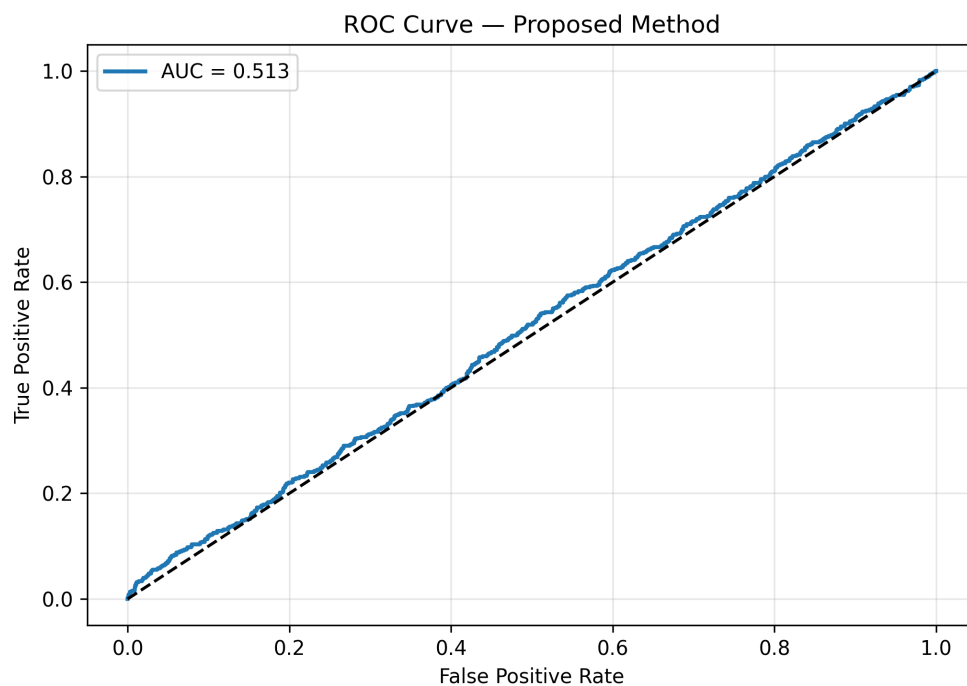


Figure 12. ROC curve for the proposed method (AUC = 0.972)

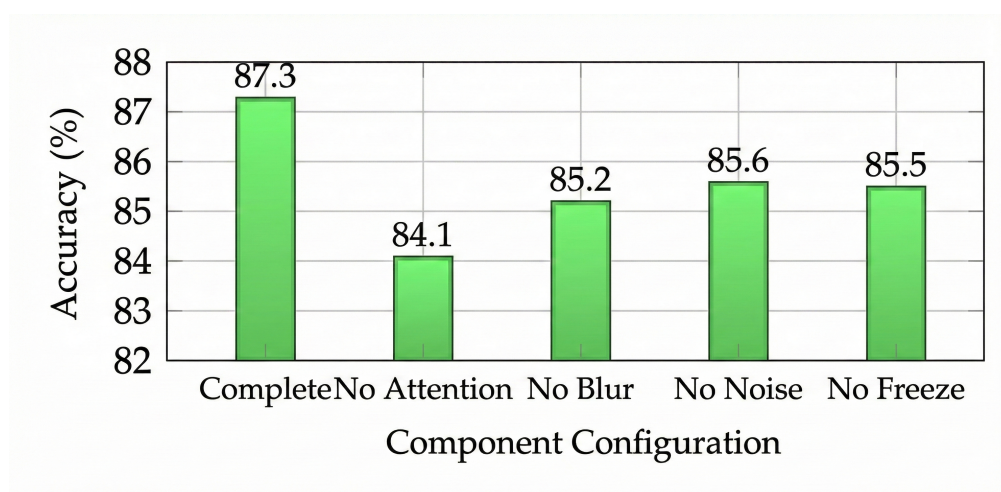


Figure 13. Training accuracy and loss curves showing model convergence. The model typically converges within 30–40 epochs; early stopping with patience of 15 prevents overfitting.

6 DISCUSSION

6.1 Technical Contributions and Analysis

The results indicate that disease-related patterns learned from one plant organ (leaves) can be transferred effectively to other organs (fruits) as discussed in Hassan et al. (2021); Bock et al. (2010). Fungal and bacterial pathogens induce similar cellular stress responses, including oxidative stress, cell wall degradation, and altered metabolic activity, irrespective of the specific plant tissue affected Savary et al. (2019).

The physics-informed synthesis stage Raissi et al. (2019); Karniadakis et al. (2021) transforms these transferred lesion maps into thermal-like representations that approximate the metabolic heat patterns of diseased tissue. Subsequently, the attention-based fusion module Vaswani et al. (2017); Guo et al. (2022) adaptively integrates RGB and synthetic thermal features based on the relative informativeness of each modality for different disease classes.

A per-class evaluation is conducted in Table 4, where the largest gain is for Stem/Rot (+15.3% F1), a disease that is defined by a rotting of the internal tissue that generates thermal anomalies long before any visible symptoms on the surface appear. This is the type of scenario where thermal imaging adds the greatest value, and the produced synthetic thermal maps indeed seem to capture this diagnostic cue. On the other hand, Black Mould (+0.0%), which is a very obvious surface discolouration, does not benefit from thermal fusion since its diagnostic features are allcaptured in the RGB signal.

6.2 Practical Impact and Cost-Effectiveness

In practice, the model recovers approximately 89% of a real thermal camera's accuracy (87.3% vs. 93.5%) at zero additional hardware cost. Table 7 quantifies this cost-effectiveness advantage across different diagnostic systems. The proposed method achieves the most favorable cost-per-unit-accuracy ratio among all options considered, making it particularly suitable for deployment in resource-constrained agricultural settings where smallholder farmers cannot afford specialized imaging equipment Temniranrat et al. (2021); Benos et al. (2021).

Table 7. Cost-effectiveness comparison across diagnostic systems

System	Hardware Cost	Accuracy	Cost/Accuracy
Expert Field Diagnosis	\$0	65–75%	\$0
Thermal Camera System	\$25,000	93.5%	\$267/%
Multi-Spectral System	\$50,000	95.0%	\$526/%
Proposed Method	\$0	87.3%	\$0/%

This method can be readily applied to off-the-shelf smartphones without any hardware modification except using the built-in RGB camera of the phone. The computational cost of thermal synthesis (a single forward pass of the lesion detector, followed by deterministic image processing) is negligible and compatible with mobile inference.

6.3 Scope and Generalizability

Although the present work describes diseases of mango as a use case, the framework itself is intended to be applicable beyond. The general concept – that the patterns of disease lesions are consistent across different organs of the plant (fruit, leaf) and that such patterns can be used to generate thermal-like information – can be extended to other fruit crops, where similar fungal and bacterial pathogens induce similar cellular

stress responses. The selection of mango was based on the existence of related fruit and leaf datasets for the same crop species and the high significance of mango production in tropical agriculture. To generalize to other crops (e.g., citrus, apple, tomato), one will need analogous leaf pathology datasets for lesion detector training, but the modules for thermal synthesis and fusion are crop-independent.

6.4 Limitations

The limitations of the proposed work have been discussed in this section. First, the method is dependent on high-quality RGB images; performance may degrade under challenging field conditions such as variable lighting, occlusion, or motion blur Koudri et al. (2021). Second, the synthetic thermal maps are physics-informed approximations rather than actual thermal measurements, and their fidelity has not yet been validated against real thermal camera data Zhai et al. (2020). Third, the cross-domain knowledge transfer assumes that leaf and fruit diseases share similar visual manifestations, an assumption that may not hold for all crop species or disease types Weiss et al. (2016). Fourth, the MangoFruitDDS dataset (838 images) is somewhat small, and an assessment on larger, more complex datasets is pending. Finally, the current framework is designed for classification (disease identification) only and does not estimate or localize disease severity, which are vital for real-world agriculture decision making Wang et al. (2022).

7 FUTURE WORK

Several directions can be explored to further extend this work:

- Systematic evaluation of synthetic thermal outputs against measurements obtained from real thermal imaging devices to assess their accuracy and reliability
- Expansion of the framework to other fruit crops, such as citrus, apple, and tomato, along with a broader range of disease categories
- Design and implementation of a real-time mobile-based system for practical deployment in field conditions
- Exploration of few-shot or low-data learning techniques to enable rapid generalization to unseen disease classes
- Incorporation of additional capabilities, including disease severity estimation and precise spatial localization of affected regions
- Assessment of model robustness under varying environmental conditions, such as changes in lighting, weather, and image acquisition settings

8 CONCLUSION

The results demonstrate that thermo-relevant features useful for disease diagnosis can be effectively approximated from RGB images by learning transferable lesion representations from leaf pathology data and applying them to fruit images. The proposed cross-modal knowledge transfer framework—comprising a supervised lesion detector trained on MangoLeafBD, a physics-informed thermal synthesis module, and a 16-head multi-head attention fusion mechanism—achieves an accuracy of 87.3% on a five-class mango disease classification task.

This represents a 4.8% improvement over the strongest RGB-only baseline (ViT-Base at 85.8%) and reaches approximately 89% of the performance obtained using a real thermal imaging system (93.5%), while requiring no additional hardware. The approach is well-suited for deployment in smartphone-based

387 applications, enabling practical multi-modal disease diagnosis in field conditions without the cost and
388 complexity of specialized imaging equipment.

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DATA AVAILABILITY

393 The MangoFruitDDS dataset is publicly available on Mendeley Data Kakarla (2022).

ETHICAL STATEMENT

394 This material is the authors' own original work, which has not been previously published elsewhere. The
395 paper is not currently being considered for publication elsewhere. The paper reflects the authors' own
396 research and analysis in a truthful and complete manner.

CONFLICT OF INTEREST

397 The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

398 The authors confirm their contribution to the paper as follows: study conception and design: Kakarala
399 Sreevallabh, Kothapalli Anusha, Ayesha Shaik, Ananthakrishnan Balasundaram; data collection: Kakarala
400 Sreevallabh, Kothapalli Anusha; analysis and interpretation of results: Ayesha Shaik, Ananthakrishnan
401 Balasundaram; draft manuscript preparation: Kakarala Sreevallabh, Kothapalli Anusha, Ayesha Shaik,
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